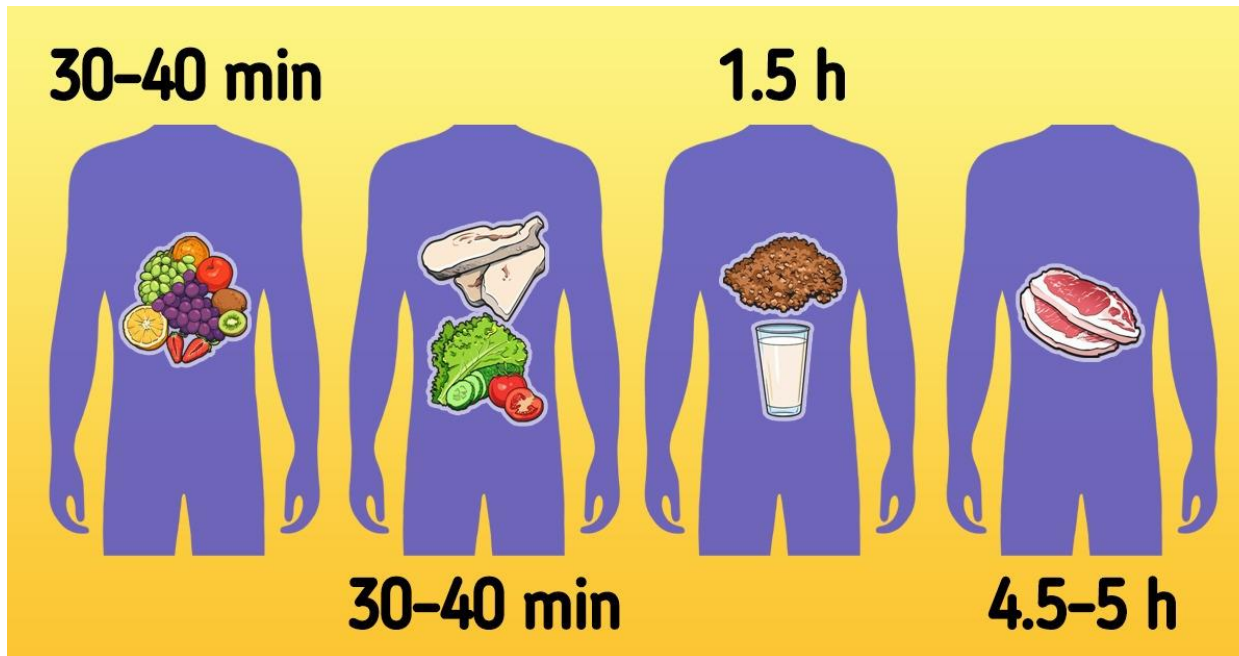
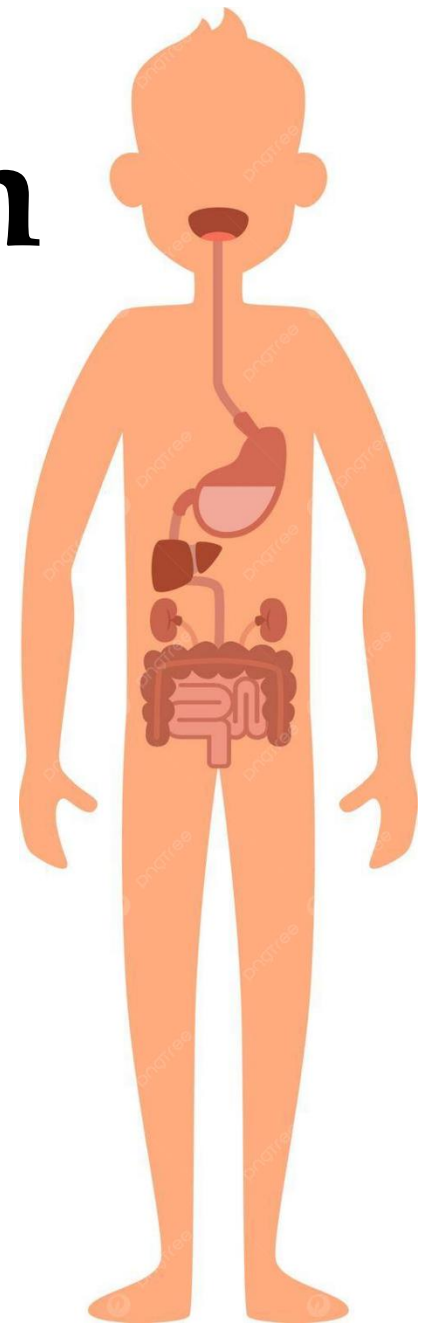


Digestive System

Lecture 19



Different diets; different lives

- All animals eat other organisms

- Herbivores

- eat mainly plants
 - gorillas, cows, rabbits, snails

- Carnivores

- eat other animals
 - sharks, hawks, spiders, snakes

- Omnivores

- eat animals & plants
 - cockroaches, bears, humans
 - humans evolved as hunters, scavengers & gatherers

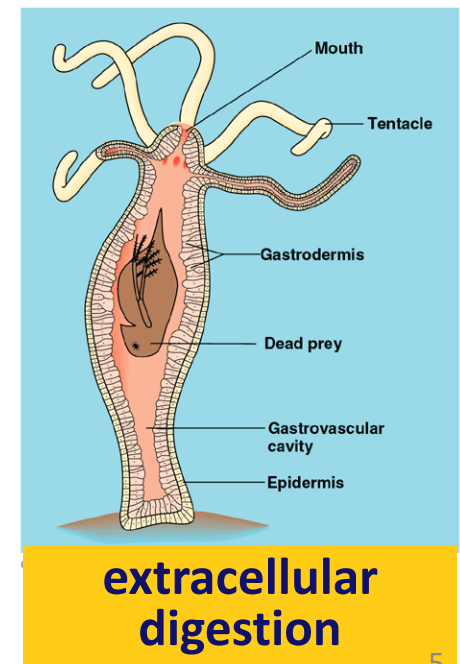
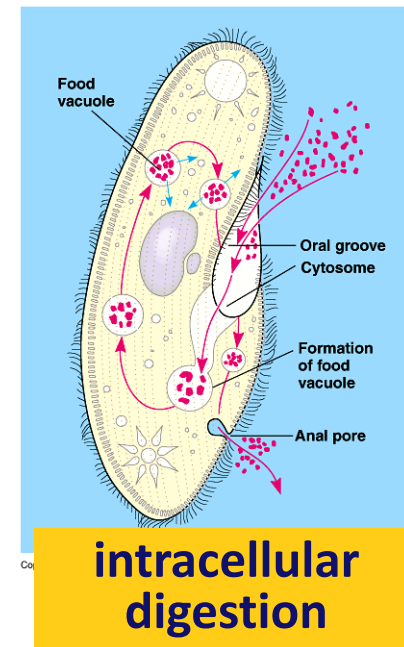


Getting & Using Food

There are several stages of **holozoic**

- **Ingest** **nutrition**:
 - taking in food
- **Digest**
 - mechanical digestion
 - breaking up food into smaller pieces
 - chemical digestion
 - breaking down food into molecules small enough to be absorbed into cells
 - enzymes
- **Absorb**
 - absorb nutrients across cell membranes
 - diffusion
 - active transport
- **Eliminate**
 - undigested material passes out of body

Holophytic nutrition - *photosynthesis or chemosynthesis*, and
Saprozoic nutrition - enzymes are released externally



Components of the Digestive System

The Human Digestive System:

- a) the alimentary canal or gastrointestinal tract
- b) several accessory organs

The alimentary canal or gastrointestinal tract (GI tract):

A tube, or gut, through which food travels in one direction.

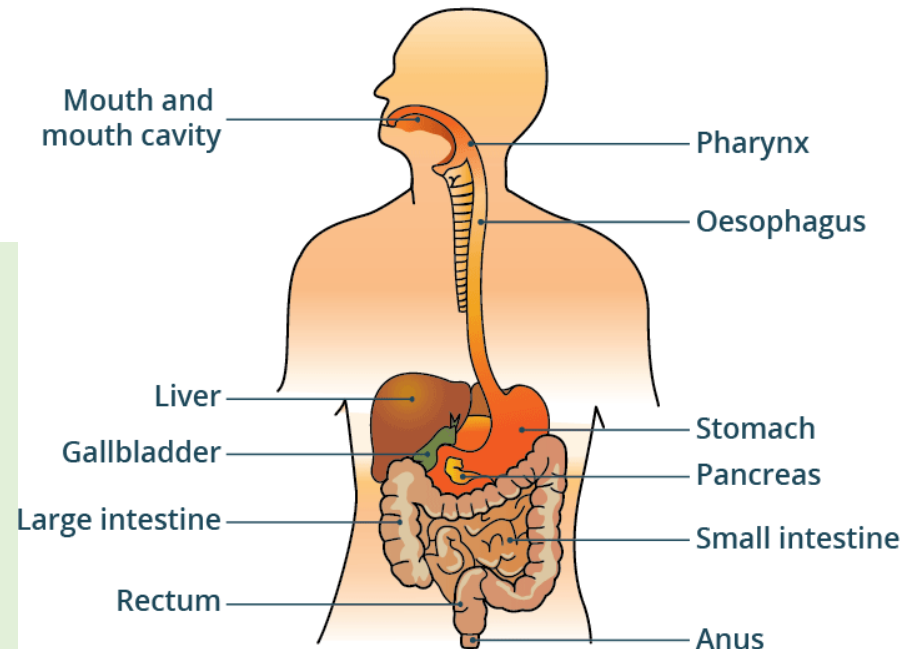
Components:

1. Mouth
2. Oesophagus
3. Stomach
4. Small intestine
5. Large intestine
6. Rectum
7. Anus

b) Accessory organs:

Aid in digestion.
However, food does not pass through them.

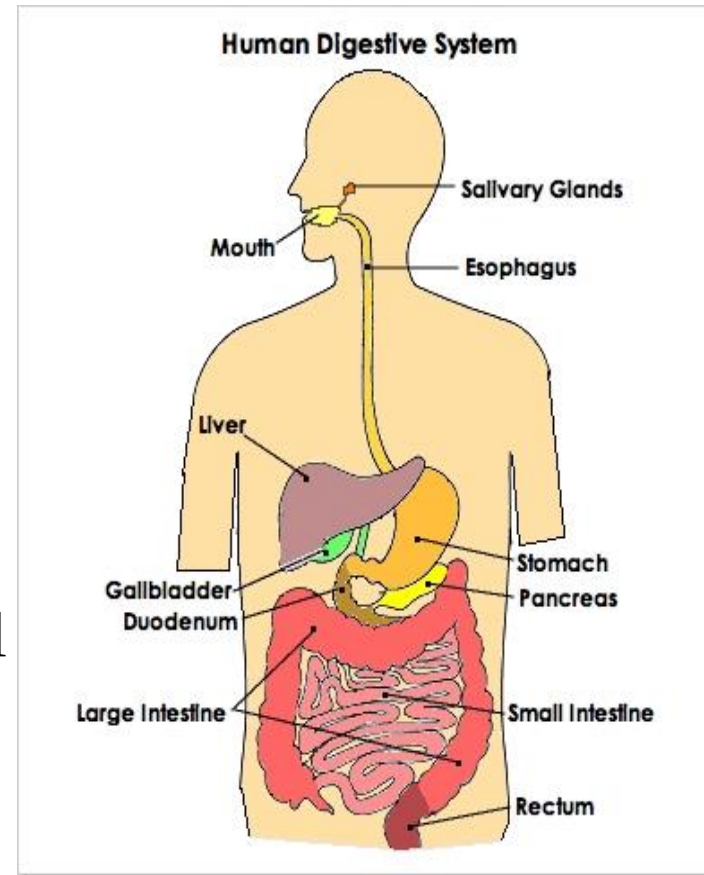
- Liver
- Gall bladder
- Pancreas



Human Digestive System

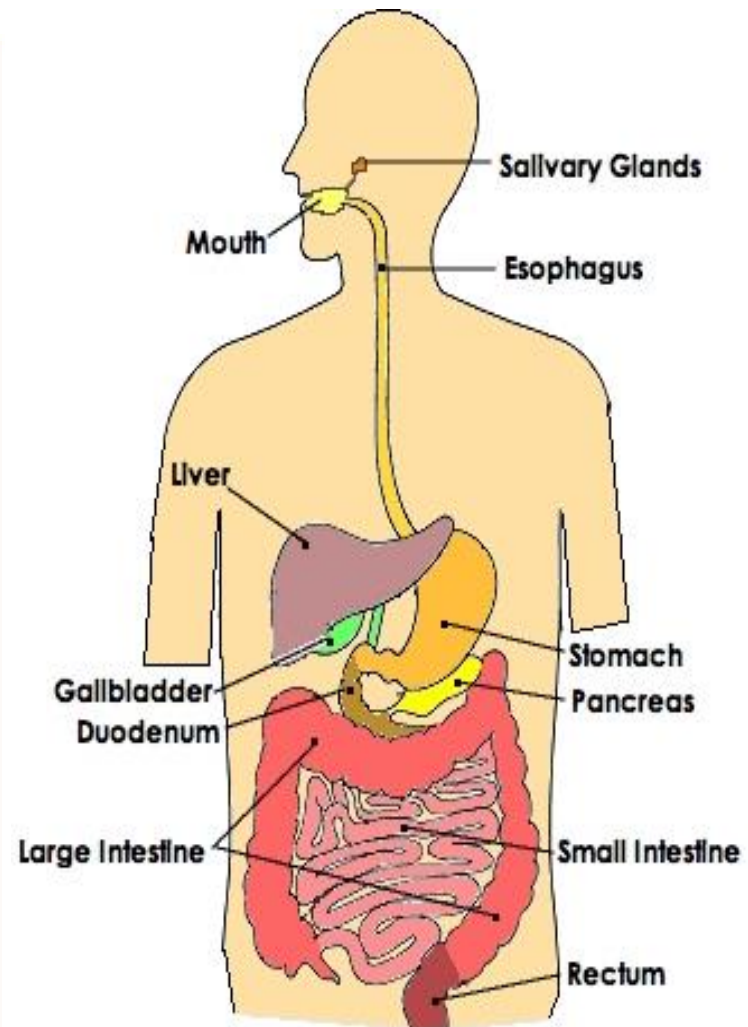
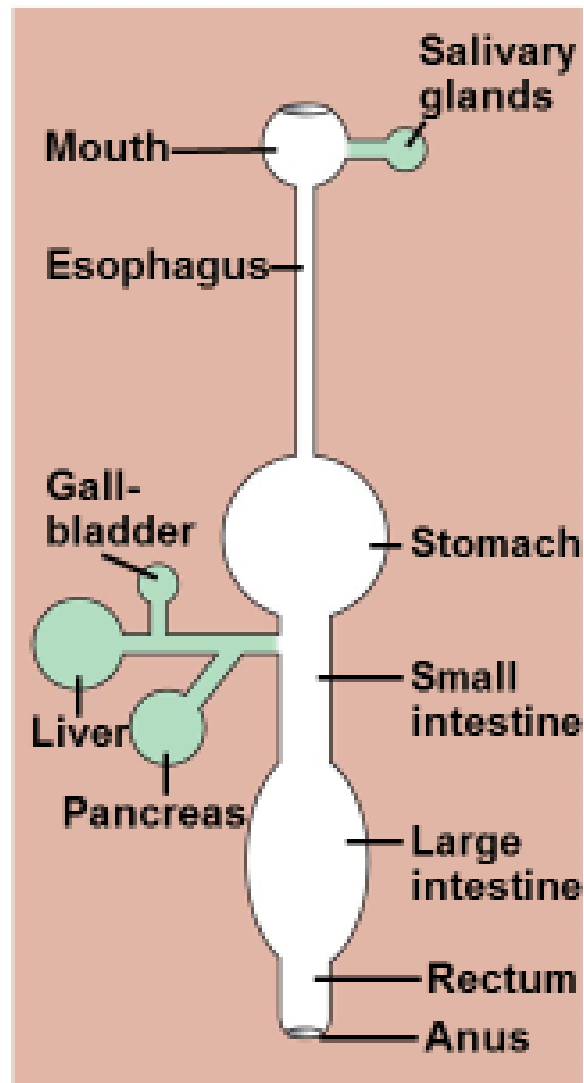
The human alimentary canal:

- The alimentary canal begins at the **mouth** and the **pharynx**.
- From here, food travels down a muscular tube called the **esophagus** and into the **stomach**.
- The stomach connects to the **small intestine**, which in turn connects to the **large intestine**.
- Some form of mechanical and chemical digestion breaks down food at every step along this tube and,
- Nutrients are absorbed across the walls of both intestines.
- All products remaining after food travels through the large intestine are waste, which exit the body through the **anus**.



GI Tract

A Schematic Diagram of the Human Digestive System



Oral cavity

The oral cavity (Digestion in Mouth):

Mouth: teeth, tongue, salivary glands.

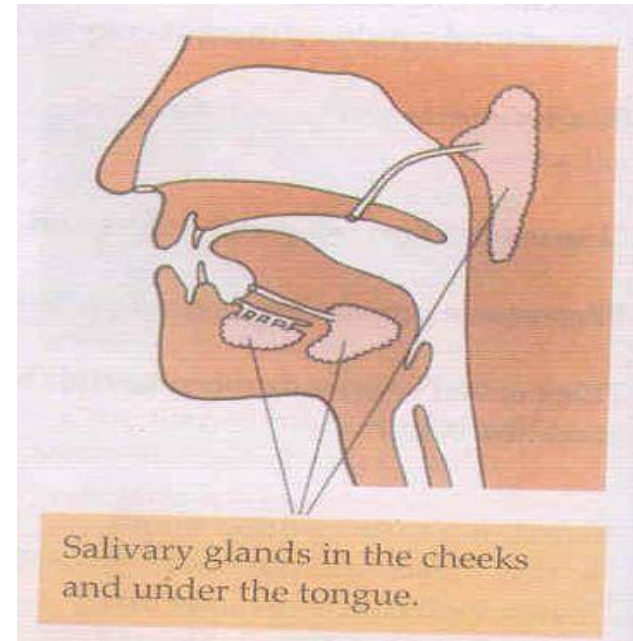
Teeth: break food into smaller pieces to increase surface area for the digestive enzymes to work on.

Tongue: a muscle used to manipulate food and form it into a **bolus**, a round, easy-to swallow ball.

Salivary glands: secrete mucous **saliva**

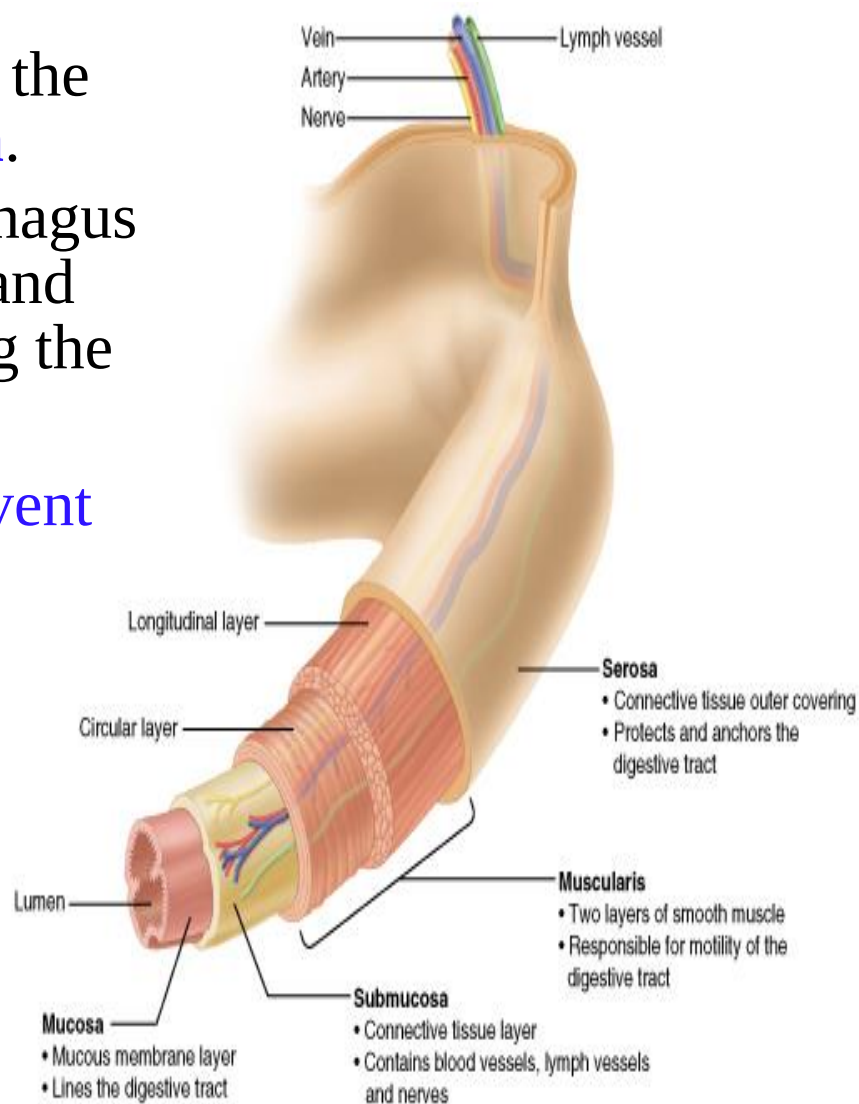
Saliva:

- contains water, salt and an enzyme called **salivary amylase** that breaks down starch into glucose.
- contains **buffers**, substances that neutralize acidic foods.
- contains **antibacterial substances** that kill bacteria in the food.
- also eases the **passage** of food through the **pharynx** and **esophagus**.

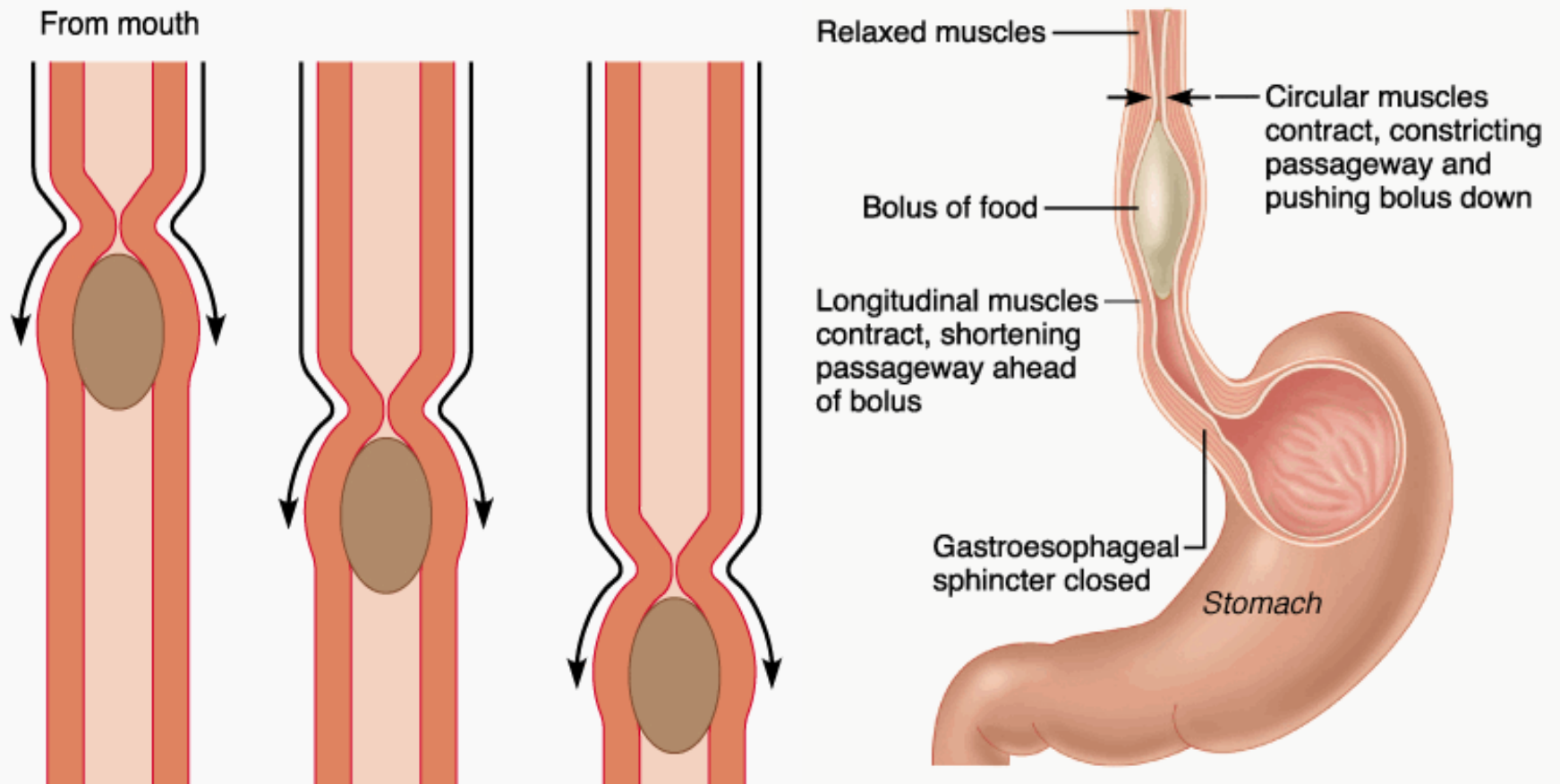


Esophagus

- The esophagus **shuttles** food from the **pharynx (mouth) to the stomach**.
- The muscles surrounding the esophagus perform **peristalsis**, the rhythmic and stepwise contraction to move along the esophageal passage.
- Sphincter muscles at each end **prevent backflow**.
- The wall of the digestive tube is composed of **4 basic layers**:
Serosa, **Muscularis**, **Submucosa** and **Mucosa**.



Bolus



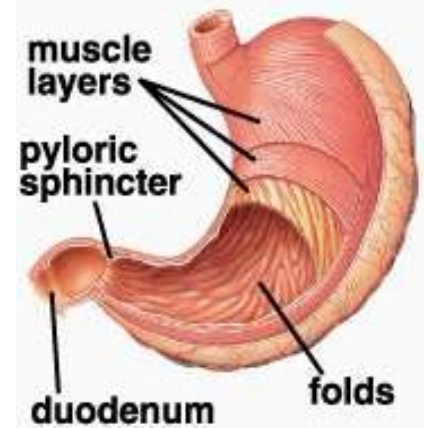
Stomach

It is a sac with inner surface is highly convoluted (extremely folded), allowing to fold up and expand.

- Digestion occurs in the stomach as gastric juice, is secreted by gastric glands located in the stomach lining.
- Gastric juice is an acidic mixture of enzymes and mucus.

Gastric glands are composed of two types of cells:

1. **Parietal cells:** secrete hydrochloric acid (HCl)
 2. **Chief cells:** secrete pepsinogen, precursor to the enzyme pepsin.
- Protein digestion is initiated by the acidic environment (pH 2) of the stomach.



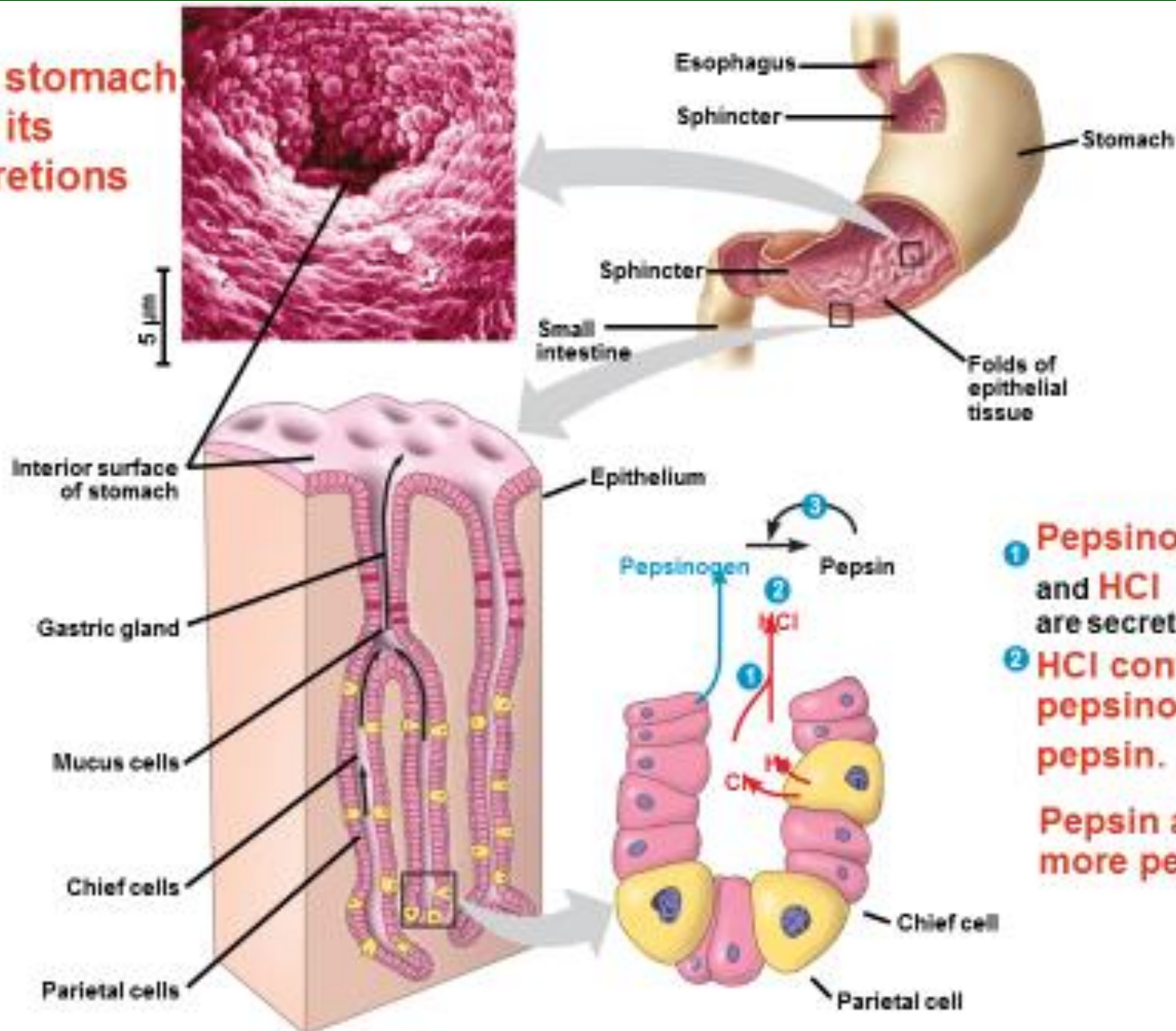
Structure and functions of components of Human Digestive System

The Stomach (Contd.)

- Low pH levels are necessary for pepsinogen to be converted into the active digestive enzyme pepsin, which further breaks down food proteins.
- If chief cells were to secrete pepsin directly into the stomach, the pepsin would digest the cells themselves.
- Additionally, mucus lines the stomach walls and prevents it from being digested.

A mixture of gastric acid and partially digested food called chyme passes out of the stomach.

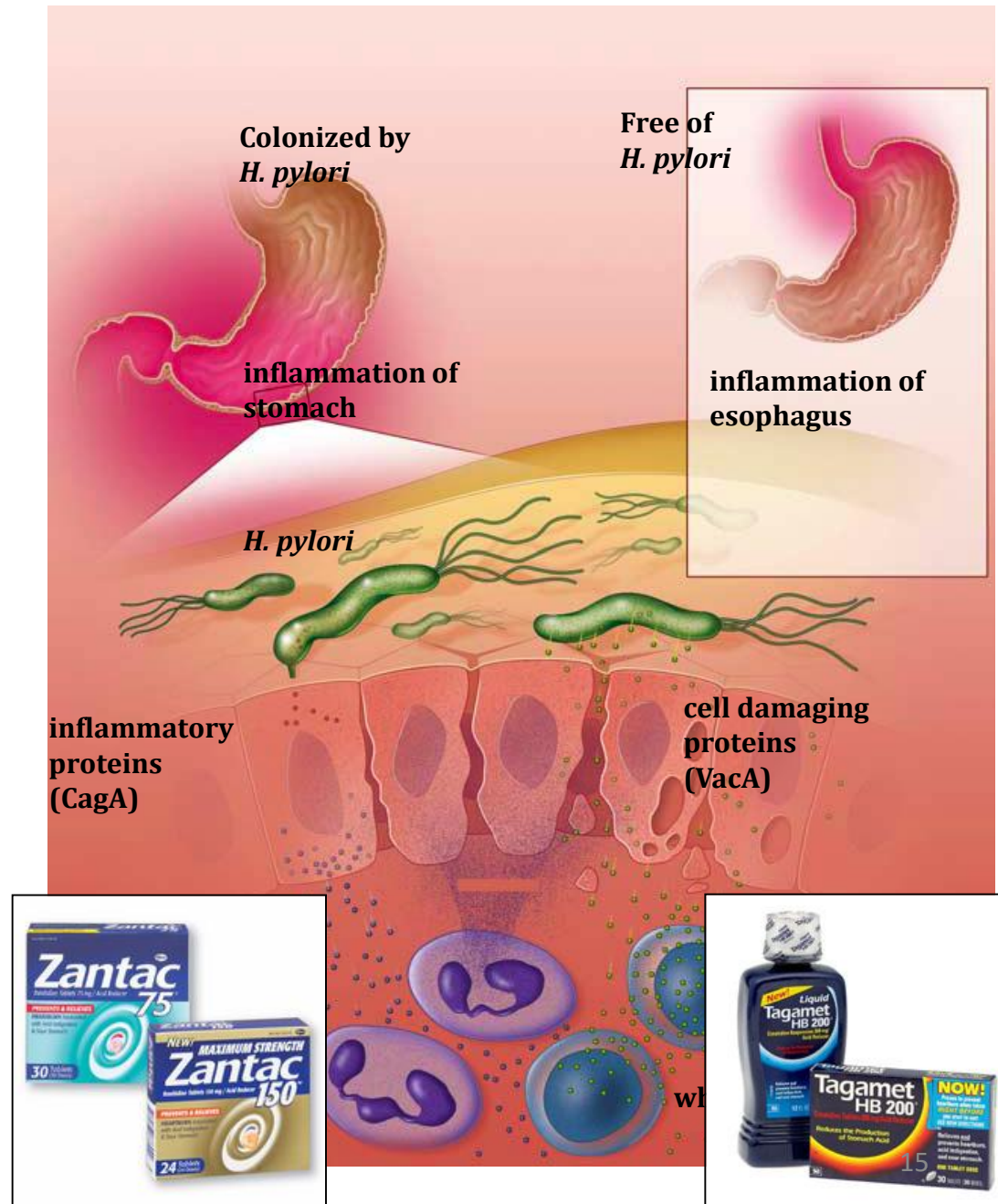
The stomach and its secretions



- 1 **Pepsinogen and HCl are secreted.**
- 2 **HCl converts pepsinogen to pepsin.**
- 3 **Pepsin activates more pepsinogen.**

Ulcers

- Used to think ulcers were caused **by stress**
 - tried to control with antacids
- Now know ulcers caused by bacterial infection of stomach
 - *Helicobacter pylori* bacteria
 - now cure with antibiotics



Small Intestine

The human small intestine is over **6 m (19.6 ft)** long and is divided into three parts: the **duodenum**, the **jejunum**, and the **ileum**.

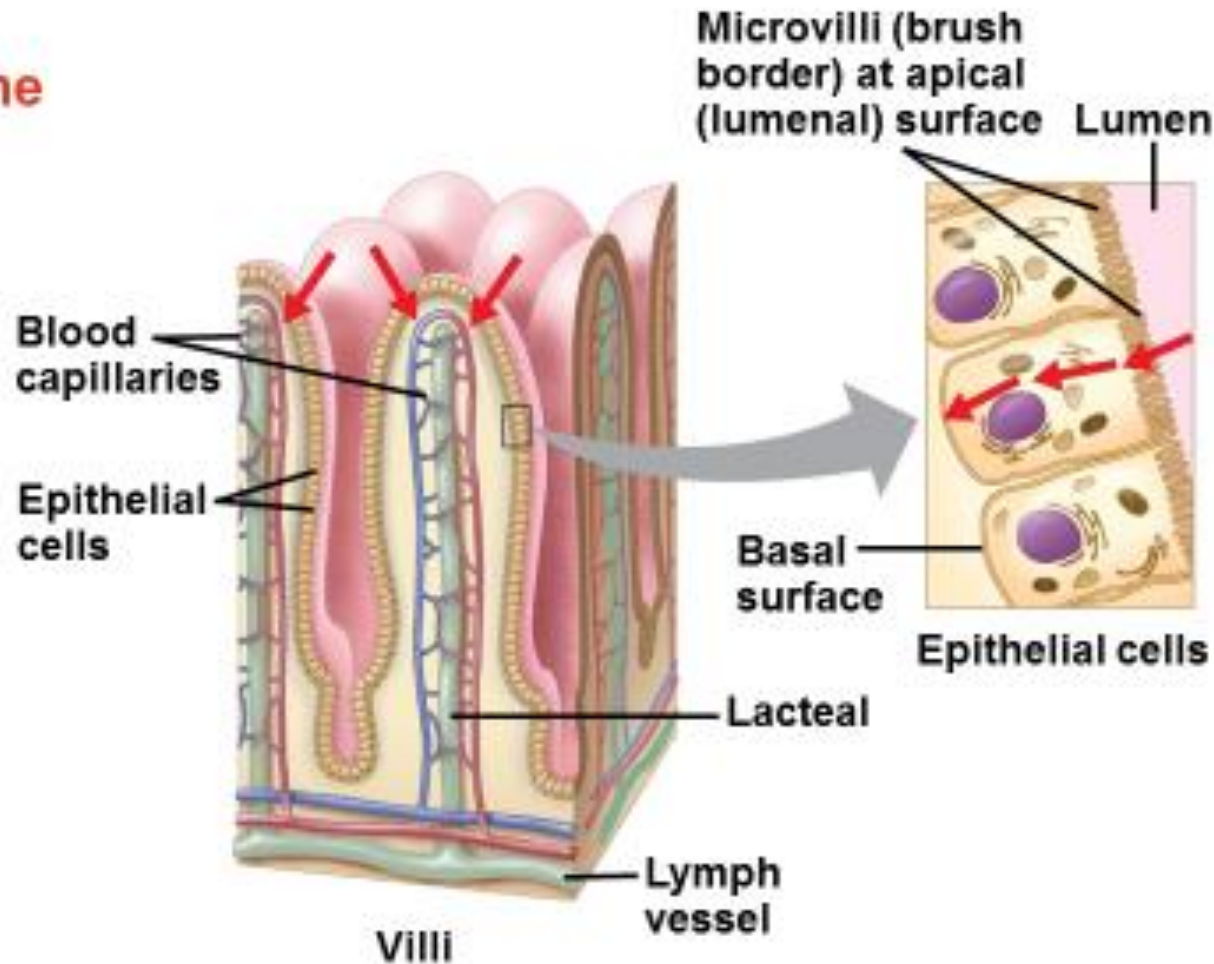
- **Duodenum** and **jejunum** → Chemical digestion

Duodenum: Controls the release of food into the small intestine

- **Ileum** → most **absorption** of the nutrients
 - Small projections called **villi** cover the ileum walls.
 - Cells lining the **villi** are covered in folds of plasma membrane that form even smaller projections called **microvilli**.
 - **Villi** and **microvilli** → increase the ileum's surface area, for absorption.
- Capillaries in the lining of the **villi** takes absorbed nutrients into the **bloodstream**, then first to the liver → rest of the body.
- Villi and microvilli projections also contain digestive enzymes to further digest food.

Structure and functions of components of Human Digestive System

Small Intestine



Key

→ Nutrient absorption

The Small Intestine

- Most of the chemical breakdown (digestion) and absorption of food takes place here.
- Bile salts break down fats in the chyme while pancreatic fluid, composed of bicarbonate, neutralizes the acid.
- The small intestine also contains a host of enzymes that help digest various food molecules.
- The enzymes present in the small intestine and their functions are outlined in the table below:

I.	Pancreatic amylase, maltase, sucrase, lactase	- Starch
II.	Trypsin, chymotrypsin, aminopeptidase, carboxypeptidase	- Protein
III.	Nucleases	- Nucleic acids
IV.	Bile salts, lipase	- Fats

Large Intestine

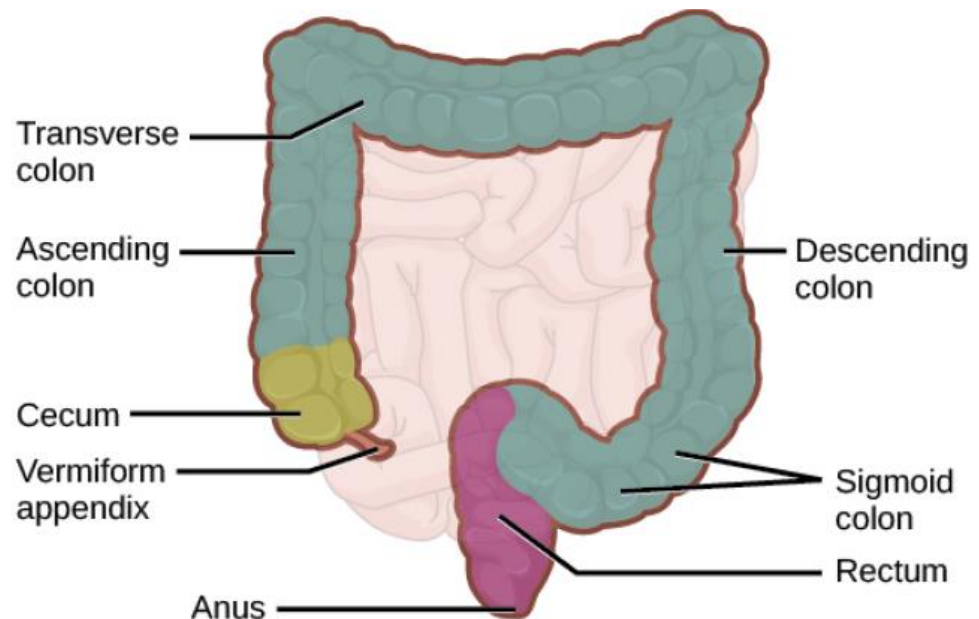
- Food from the small intestine empties into the large intestine, or **colon (1.5 meters long)**.
- Here **No digestion** and only a **small percentage of absorption**.
- It **primarily concentrate waste material into a form called feces**.
- Compact feces and move it into the **rectum**, where it is exited from the body through the anus.
- Feces is expelled through the voluntary movements of muscles surrounding the rectum.

Rectum

- Muscular storage chamber where the undigested food (feces) is held and molded before being pushed through anus during expulsion.

Anus

- The exit of alimentary canal
- It is closed by a ring of muscle (the anal sphincter)



mouth

- break up food
- digest starch
- kill germs
- moisten food

stomach

- kills germs
- break up food
- digest proteins
- store food

liver

- produces bile
 - stored in gall bladder
- break up fats

small intestines

- breakdown food
 - proteins
 - starch
 - fats
- absorb nutrients

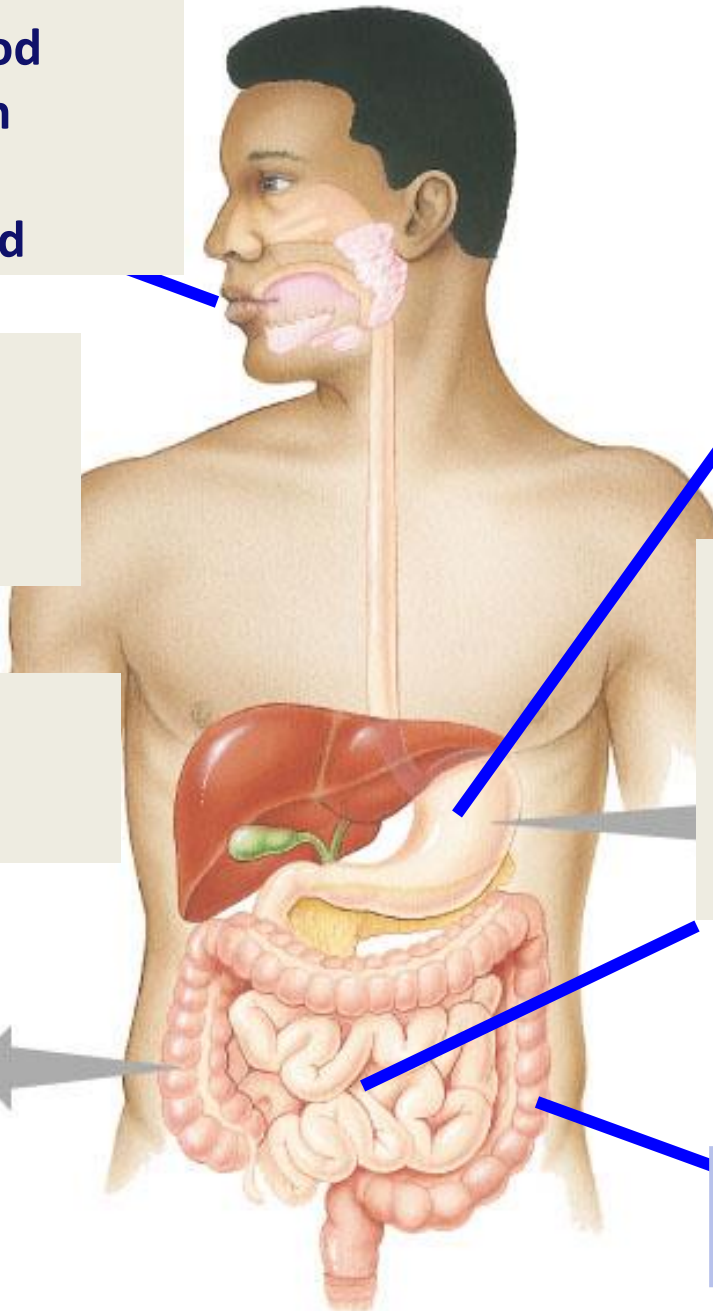
pancreas

- produces enzymes to digest proteins & carbs

large intestines

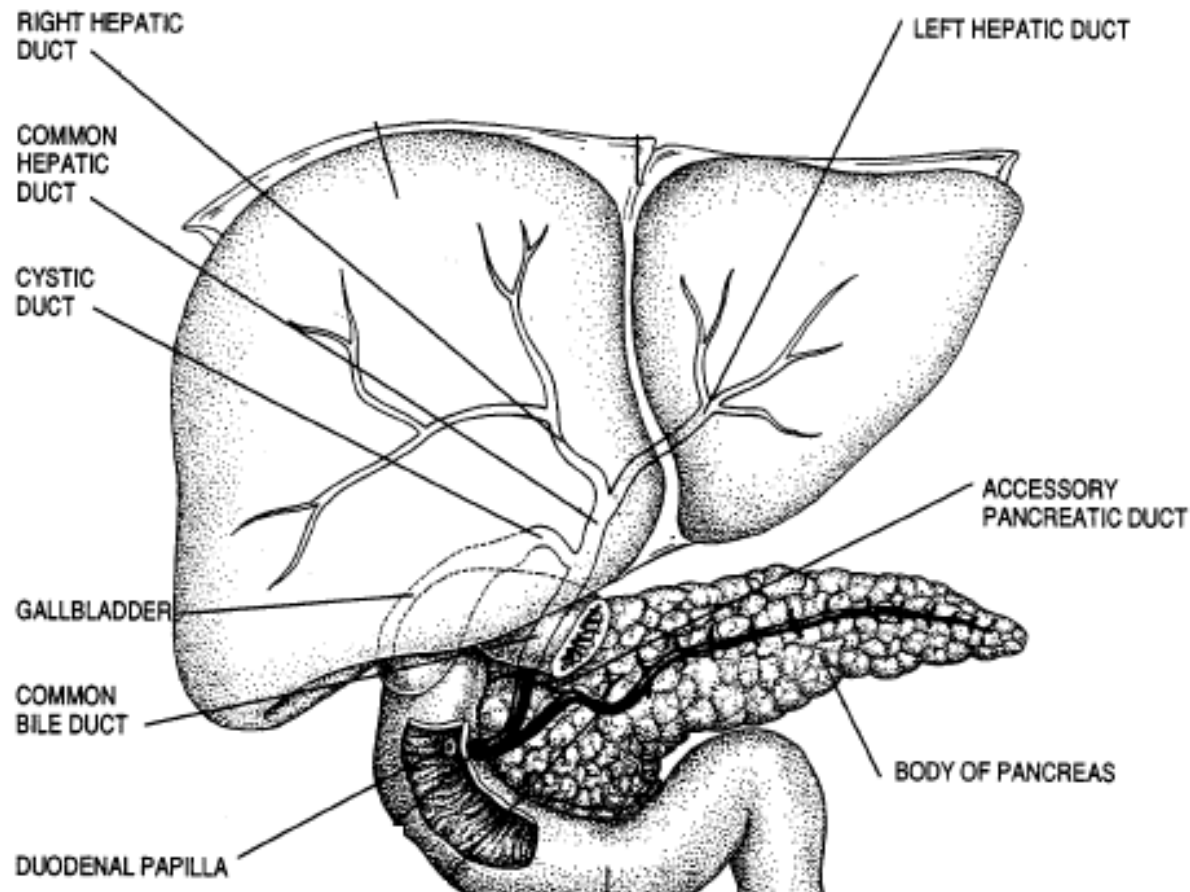
absorb water

appendix



Accessory organs

- Produce and store digestive chemicals
- Food does not pass through these

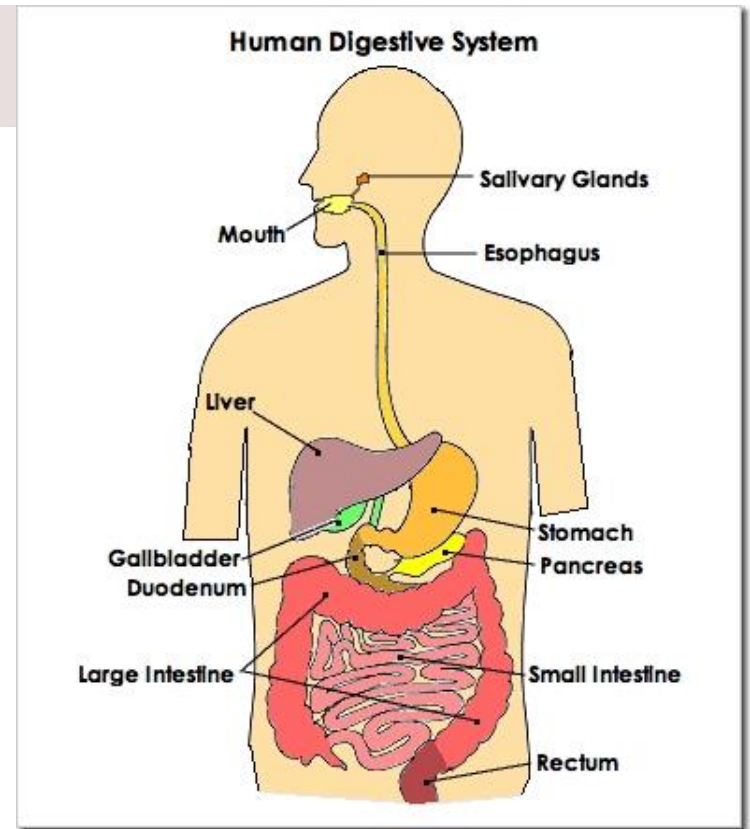


Accessory organs

- **Liver** - Produces bile, stores glucose as glycogen & releases glucose into blood stream
 - **Gallbladder** - Stores bile, and squirts bile into small intestine when fat arrives
 - **Pancreas** - Makes enzymes, empties them into small intestine
 - Pancreatic juice: **enzymes, bicarbonate**
 - Insulin and glucagon (hormones) secreted directly into blood
- Insulin:** [When glucose level is high in blood] Enables cells to use glucose (glucose enters cells) & promotes formation of glycogen in liver.
- Glucagon:** [When glucose level is low in blood] Signals the liver to convert glycogen into glucose & release it into the bloodstream.

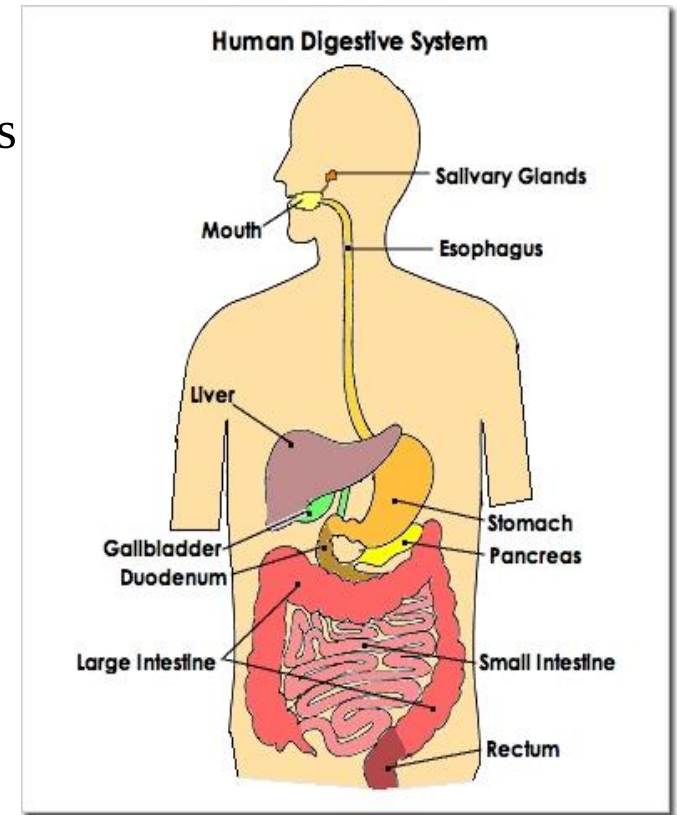
Liver

- ❖ Largest internal organ
- ❖ Called the ‘chemical factory’
- ❖ The liver has multiple functions:
 - its main function within the digestive system is to process the nutrients absorbed from the small intestine.
 - bile from the liver secreted into the small intestine also plays an important role in digesting fat.
 - the liver is the body's chemical "factory." It takes the raw materials absorbed by the intestine and makes all the various chemicals the body needs to function with.
 - the liver also **detoxifies potentially** harmful chemicals. It breaks down many drugs.



Gallbladder and Bile

- The **gallbladder** is a pear-shaped organ that stores about 50 mL of bile (or "gall") until the body needs it for digestion. It is connected to the liver and the duodenum by the biliary tract.
- **Bile** is a complex fluid containing water, electrolytes and lots of organic molecules including bile acids, cholesterol, phospholipids and bilirubin that flows through the biliary tract into the small intestine.
- Two fundamentally important functions of bile:
 - Bile contains bile acids, which are critical for digestion and absorption of fats and fat-soluble vitamins in the small intestine.
 - Many waste products, including bilirubin, are eliminated from the body by secretion into bile and elimination in feces.
- The gall bladder stores and concentrates bile during the fasting state.



Pancreas

❖ Leaf-like organ which lies between the stomach and duodenum

❖ It secretes pancreatic juice which passes to the duodenum to help in digestion

- Amylase – for digesting starch
- Protease – protein
- Lipase – fat

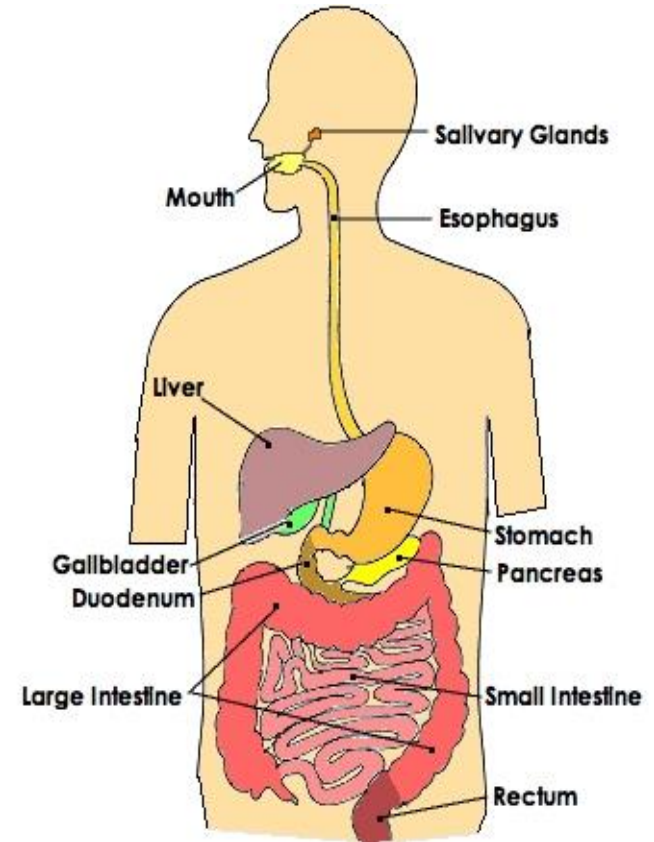
Starch $\xrightarrow{\text{amylase}}$ glucose

Proteins $\xrightarrow{\text{protease}}$ polypeptides

Fats $\xrightarrow{\text{lipase}}$ fatty acid + glycerol

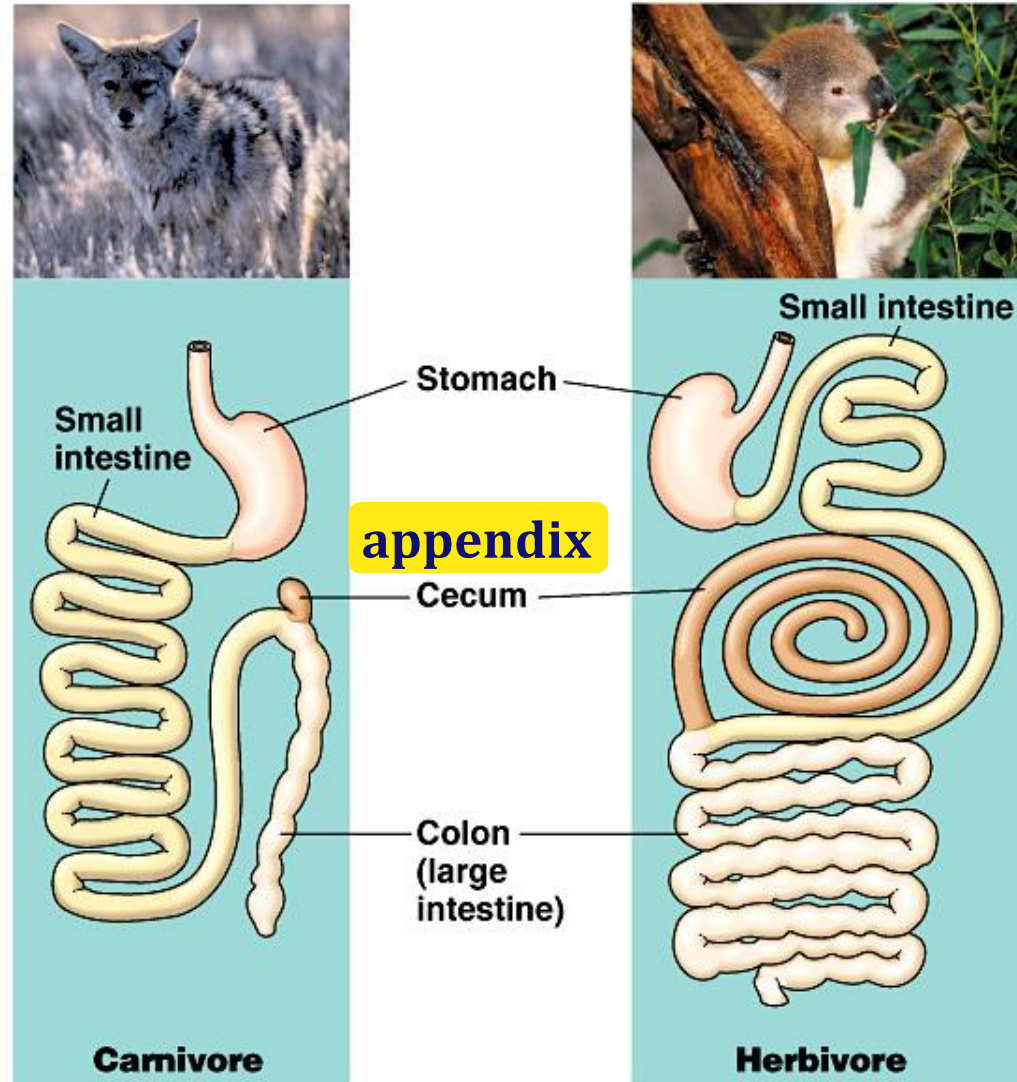
Role in Digestion:

- enzymes break down protein, fats, and carbohydrates.
- Bicarbonate in the pancreatic juice neutralizes the acidic nature of chyme from stomach.
- The pancreas also makes hormone insulin, secreting it directly into the bloodstream. Insulin is the chief hormone for metabolizing sugar.



Length of digestive system

- Herbivores & omnivores
 - **long** digestive systems
 - harder to digest cellulose (cell walls)
 - bacteria in intestines help
- Carnivores
 - **short** digestive systems
 - protein easier to digest than cellulose



Vegetarian diets

- **Need to make sure you get enough protein**
 - 20 amino acids to make protein
 - 12 amino acids humans can produce
 - 8 we have to eat = “**essential amino acids**”
 - Grains (like corn) have 6 amino acids
 - missing 2
 - Beans (like soybean & red beans) have 6 amino acids
 - missing different 2
 - mix beans & grains for complete group of amino acids
 - rice & beans
 - taco/tortilla & beans
 - tofu & rice
 - peanut butter & bread

